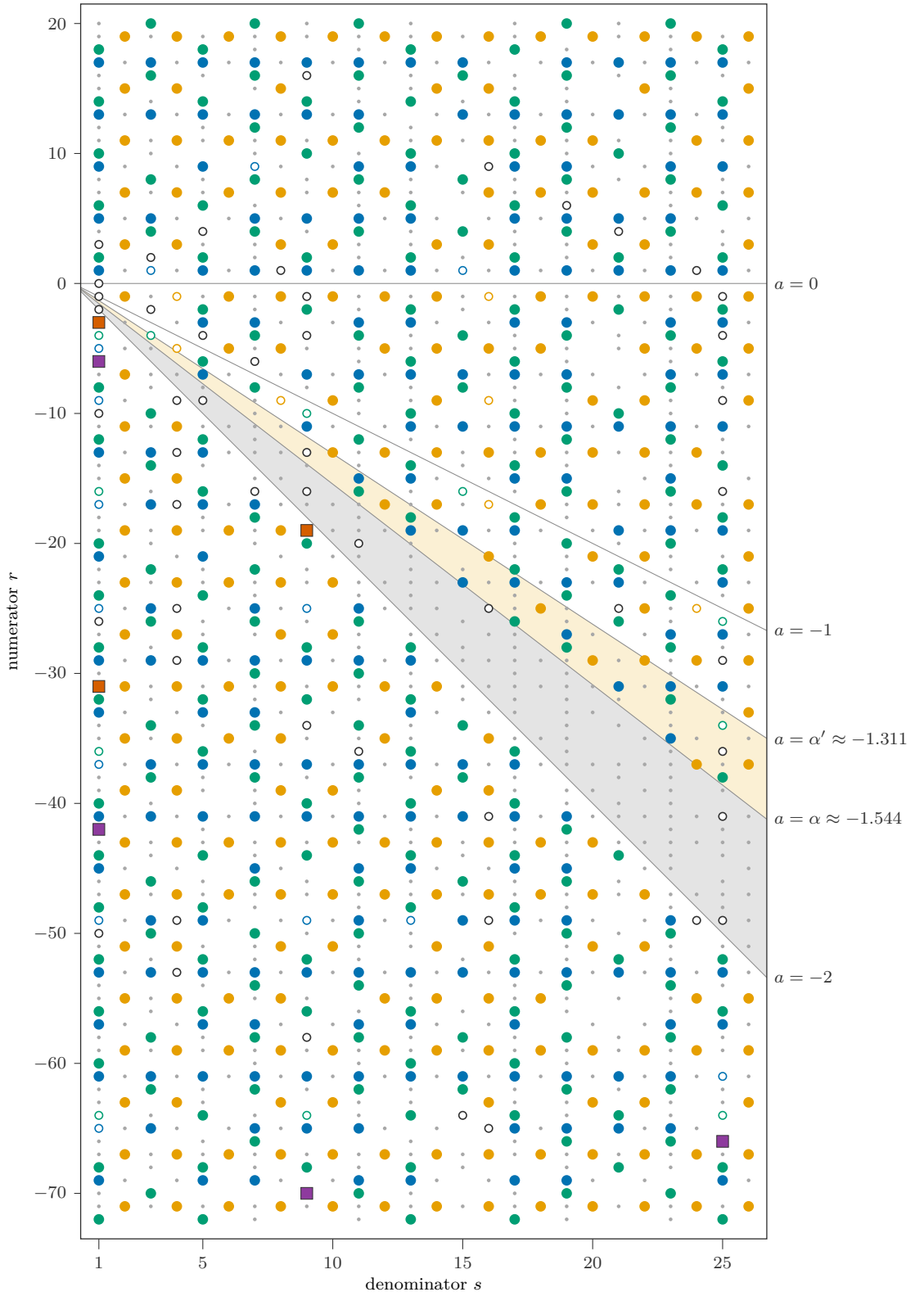



# Arboreal representations of $x^2 + a$

coprime $r, s$ with $s \geq 1$, $a = r/s$

**Main theorem: circles, by 2-adic class**

$(r, s)$ mod 4 for $a \leq -2$; for $a > \alpha$

- (C1): $(3,1)$, $(3,3)$; $(1,1)$, $(1,3)$
- (C2): $(0,1)$, $(2,3)$; $(2,1)$, $(0,3)$
- (C3): $(1,0)$, $(1,2)$; $(3,0)$, $(3,2)$

two of (Z), (O), (T) hold, so $\rho_a$ is surjective

$c_1, c_2$ are 2-dependent, so $\rho_a$ is not surjective

**The two families: squares**

$a = -\ell(\ell+1)$, $\ell = u/v > 1$, $(u, v) \equiv (2,1), (3,3) \bmod 4$

$a = -(\ell^2+\ell+1) \leq -2$, $\ell = u/v$, $(u, v) \equiv (1,1), (2,3) \bmod 4$

**Not covered**

$c_1, \ldots, c_6$ are 2-dependent, so $\rho_a$ is not surjective

$c_1, \ldots, c_6$ are 2-independent

no mark: $\gcd(r, s) > 1$; $a$ appears at its reduced fraction, whose residues mod 4 may differ

**Regions**

$-2 < a \leq \alpha$: not covered

$\alpha < a \leq \alpha'$; outside it, in the classes, $\rho_a$ is surjective iff $c_1, c_2$ are 2-independent